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Lasting immune memory against hepatitis B following challenge 10–11 years after primary vaccination with either three doses of hexavalent DTPa-HBV-IPV/Hib or monovalent hepatitis B vaccine at 3, 5 and 11–12 months of age

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ABSTRACT

Background: The combined hexavalent diphtheria–tetanus–pertussis–hepatitis B-inactivated poliomyelitis – *Haemophilus influenzae* type b conjugate vaccine (*Infanrix hexa*™; DTPa-HBV-IPV/Hib: GlaxoSmithKline Vaccines) induces robust responses to the HBV component when administered at 3, 5 and 11–12 months of age. We assessed long term HBV antibody persistence 10–11 years after primary vaccination in infancy.

Methods: Antibody persistence and immune memory were assessed post-primary vaccination at 3, 5, 11–12 months with DTPa-HBV-IPV/Hib, or monovalent HBV vaccine (*Engerix*™ B, GlaxoSmithKline Vaccines) co-administered with DTPa-IPV/Hib (*Infanrix*™-IPV/Hib, GlaxoSmithKline Vaccines) in 185 children aged 11–12 years. Blood samples were collected before and 1 month after a challenge dose of *Engerix*™ B (10 μg dose).

Results: 10–11 years after primary vaccination the percentage of subjects with persisting anti-HBs antibody concentrations ≥10 mIU/ml was 48.4% in the DTPa-HBV-IPV/Hib group and 58.4% in the DTPa-IPV/Hib + HBV group. After the HBV challenge dose, the percentage with anti-HBs ≥100 mIU/ml increased from 14.7% to 93.6% in the DTPa-HBV-IPV/Hib group and 19.1% to 94.4% in the DTPa-IPV/Hib + HBV group. Anti-HBs GMCs increased by at least 187-fold in each group. An anamnestic response (≥4-fold increase in initially seropositive or anti-HBs concentration ≥10 mIU/ml in initially seronegative subjects) was observed in 96.8% and 96.6% of subjects in the DTPa-HBV-IPV/Hib and DTPa-IPV/Hib + HBV groups, respectively. No serious adverse events occurred that were considered related to challenge vaccination.

Conclusion: Administration of HBV as part of a combination vaccine or as a monovalent vaccine induced long lasting immune memory against HBV in children primed at 3, 5 and 11 months of age. Antibody persistence and immune memory were similar, suggesting that protection afforded by DTPa-HBV-IPV/Hib and monovalent HBV vaccines, is likely to be of similar duration. The administration of HBV challenge dose 10–11 years after the 3, 5, 11–12 months primary schedule induced strong anamnestic responses and was well tolerated.

This study is registered at www.clinicaltrials.gov NCT01138098.

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Abbreviations: anti-HBs, antibody against hepatitis B surface antigen; ATP, according to protocol; CI, confidence interval; CLIA, chemiluminescence immunoassay; DTPa-HBV-IPV/Hib, hexavalent diphtheria–tetanus–pertussis–hepatitis B-inactivated poliomyelitis and *Haemophilus influenzae* type b conjugate vaccine; DTPa-IPV/Hib, diphtheria–tetanus–pertussis-B-inactivated poliomyelitis and *Haemophilus influenzae* type b conjugate vaccine; GMC, geometric mean antibody concentration; HBV, hepatitis B virus; HBsAg, hepatitis B surface antigen; SAE, serious adverse event; SD, standard deviation; WHO, World Health Organization.

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1. Introduction

The World Health Organization (WHO) estimates that more than 2 billion individuals have been infected with hepatitis B virus (HBV) globally, of whom 360 million suffer chronic infection with its consequent risks of transmission and premature death due to cirrhosis or hepatocellular carcinoma [1]. Hepatitis B is most likely to become chronic when infection occurs during infancy and early childhood [1–3]. In countries of low HBV endemicity a substantial proportion of chronic disease cases arise through perinatal or early childhood infections [4]. Therefore, in these countries, primary vaccination commencing in infancy offers the most effective means by which high coverage can be achieved, thereby preventing HBV infection throughout life. Vaccination against HBV commencing at birth is the primary preventative strategy adopted by the WHO and is recommended by the Viral Hepatitis Prevention Board [5,6].

Infanrix hexa™ is a combined hexavalent diphtheria–tetanus–pertussis–HBV-inactivated poliomyelitis and *Haemophilus influenzae* type b conjugate vaccine (DTPa-HBV-IPV/Hib, GlaxoSmithKline Vaccines, Wavre, Belgium). The immunogenicity and clinically acceptable safety profile of DTPa-HBV-IPV/Hib when used for primary vaccination of infants and as a booster dose in the second year of life has been extensively documented in clinical trials [7]. DTPa-HBV-IPV/Hib may be used as a 3-dose primary vaccination course with a booster in the second year of life, or as a 2-dose primary and booster regimen at 3, 5 and 11–12 months of age, as employed by Denmark, Iceland, Sweden, Finland, Italy, Norway and Austria [8]. The use of combined vaccines including DTPa-HBV-IPV/Hib in infant vaccination schedules has several advantages, including improved timeliness of vaccination and improved coverage of recommended vaccine antigens [9,10].

The HBV component in DTPa-HBV-IPV/Hib induces robust responses with at least 97.5% of subjects achieving anti-HBsAg (anti-HBs) antibody concentrations ≥10 mIU/ml after 3 primary vaccination doses [7]. In children vaccinated with 2 doses of DTPa-HBV-IPV/Hib at 3 and 5 months of age, 96.8% had anti-HBs antibodies ≥10 mIU/ml, increasing to 98.9% after the third dose at 11–12 months of age [11]. These results are similar to those observed when monovalent HBV is co-administered with DTPa-IPV/Hib at 3, 5 and 11–12 months [12].

Recent studies of antibody persistence show that by 7–9 years of age, 78% of children continue to have anti-HBs antibody concentrations ≥10 mIU/ml after four doses of DTPa-HBV-IPV/Hib, administered during infancy and the second year of life [13–15]. Furthermore, anamnestic responses were demonstrated on re-exposure to HBV in the majority of subjects who had become seronegative [13,15]. These findings are consistent with the observation that primary vaccination induces long lasting immune memory which is present even in individuals without detectable anti-HBs antibodies many years after vaccination [16,17]. Because current data indicate that maintenance of detectable anti-HBs antibody levels is not required for ongoing protection against HBV disease, the WHO and the EU Consensus Group on Hepatitis B Immunity currently advise that no HBV booster dose is required after 3-dose primary vaccination in immunologically competent individuals [1,17,18].

We evaluated antibody persistence 10–11 years after vaccination at 3, 5, 11–12 months with DTPa-HBV-IPV/Hib or monovalent HBV vaccine (*Engerix*™ B, GlaxoSmithKline Vaccines) co-administered with *Infanrix*™-IPV/Hib (DTPa-IPV/Hib, GlaxoSmithKline Vaccines) [12], and assessed the ability of these children to mount an anamnestic response to a challenge dose of HBV vaccine at year 10–11.

2. Methods

Results of the primary vaccination study (study 031) conducted in 1998–1999 have been reported previously [12]. The phase 4 open, follow up and challenge study was conducted in 12 centres in Slovakia between 07 June 2010 and 26 November 2010 (113954, www.clinicaltrials.gov NCT01138098) according to Good Clinical Practice guidelines and the Declaration of Helsinki. The study protocol was approved by local ethics committees. Written informed consent was obtained from parents/guardians and written informed assent was obtained from subjects depending on their level of comprehension.

All 312 children who participated in the previous vaccination study 031 were invited to participate in the follow-up study. Subjects were not eligible to participate if they had received HBV vaccination since completing the initial vaccination study; if they had a history of HBV infection; if they had received immunoglobulin or blood products within the previous 3 months or planned to receive them during the study period; if they were immunosuppressed from any cause including chronic use of immunosuppressants; if they had received/planned to receive any other vaccine within 30 days before or after the HBV dose. Females who were pregnant, lactating or planning to become pregnant were also unable to participate.

Participants received 3 doses of either DTPa-HBV-IPV/Hib, or DTPa-IPV/Hib + HBV, at 3, 5, 11 months of age in study 031. Children retained the same group allocation that they were randomized to in the initial vaccination study. In this follow-up study, all children (groups DTPa-HBV-IPV/Hib and DTPa-IPV/Hib + HBV) received a single 0.5 ml challenge dose of *Engerix*™ B (10 μg HBsAg; Lot number: AHBVB720A) administered intramuscularly in the non-dominant deltoid using a 25 gauge needle measuring at least 2.54 cm in length.

2.1. Immunogenicity assessment

Blood samples were collected prior to, and 1 month post-challenge from all subjects. Anti-HBs antibodies were measured using an FDA approved and EU marketed chemiluminescence immunoassay (CLIA) (Centaur™, Siemens, Germany) with a cut-off of 6.2 mIU/ml defining seropositivity and 10 mIU/ml defining seroprotection.

2.2. Safety and reactogenicity assessment of the challenge dose

Participants were asked to record any solicited local and general adverse events that occurred during 4-day (Day 0–3) follow-up after the challenge dose. Serious adverse events (SAEs) were recorded over the entire study period.

2.3. Statistical analysis

Three study cohorts were used for analysis: the According to Protocol (ATP) Persistence Cohort included all subjects who had received 3 vaccine doses in the primary study 031, who had not received any additional dose of HBV after the primary vaccination, who had no evidence of HBV infection or disease, and who had pre-challenge blood sampling results available. The Total Vaccinated Cohort included all subjects who received the HBV challenge dose. The ATP Immunogenicity Cohort included all subjects who received the challenge dose, who met eligibility criteria, complied with protocol-defined procedures and who had post-challenge blood sampling results available. Subjects were eliminated from the ATP Immunogenicity Cohort if they had serological evidence of previous HBV infection. (anti-hepatitis B core antigen positive or an abnormal increase in anti-HBs antibody concentration since completion

of study 031. An abnormal increase was defined as a pre-challenge anti-HBs antibody concentration greater than the post-dose 3 antibody concentration in study 031.) In the present study the ATP immunogenicity cohort and the ATP persistence cohort were identical, and are referred to as the ATP immunogenicity cohort.

The primary objective of the study was to assess the percentage of subjects who achieved anti-HBs antibody concentrations ≥100 mIU/ml after the challenge dose. Because 10–11 years had elapsed since the initial vaccination study, we were unable to estimate how many subjects would be available or willing to participate in this follow-up study. All the subjects in the primary study were invited to participate in the challenge phase. Assuming that 10–15% subjects might not evaluable at the time of analysis, at best 270 subjects might contribute to the ATP analyses. The bounds of the exact 95% confidence intervals (CIs), if 88% of subjects achieved anti-HBs antibodies ≥100 mIU/ml would be [83.7%; 91.8%] when the sample size was 270.

Secondary objectives of the study included: the assessment of antibody persistence 10–11 years after vaccination; assessment of anamnestic responses measured 1 month after the challenge dose (defined as a 4-fold or greater increase in the post-challenge anti-HBs antibody concentration in subjects who were seropositive prior to the challenge dose; or a post-challenge anti-HBs antibody concentration ≥10 mIU/ml in initially seronegative subjects); and assessment of the reactogenicity and safety of the challenge dose. The association between pre- and post-challenge responses was explored graphically.

3. Results

One hundred eighty-five subjects between 11 and 12 years of age returned for the follow-up visit at year 10–11 (Fig. 1). Only one subject was eliminated from the ATP Persistence and Immunogenicity Cohorts because they had serological evidence suggesting either HBV infection or additional vaccination. Subjects in both groups had a mean age of 11.3 years at the follow-up visit (range 11–12 years). The percentage of subjects who were female was 47.4% (45/95) in the DTPa-HBV-IPV/Hib group and 38.2% (34/89) in the DTPa-IPV/Hib+HBV group, which is consistent with the gender distribution of the initial primary vaccination study (50.6% female in the DTPa-HBV-IPV/Hib group and 42.9% in the DTPa-IPV/Hib+HBV group).

3.1. Antibody persistence and immunogenicity of the challenge dose

Ten to 11 years after primary and booster vaccination in infancy at 3, 5 and 11–12 months of age, 55.8% (95% CI 45.2%; 66.0%) of subjects in the DTPa-HBV-IPV/Hib group and 64.0% (53.2%; 73.9%) in the DTPa-IPV/Hib+HBV group continued to have detectable anti-HBs in serum (≥6.2 mIU/ml) (Table 1). Seroprotective concentrations (anti-HBs antibody concentrations ≥10 mIU/ml) were present in 48.4% (95% CI 38.0%; 58.9%) of subjects in the DTPa-HBV-IPV/Hib group and 58.4% (95% CI 47.5%; 68.8%) in the DTPa-IPV/Hib+HBV group (Table 1).

The percentage of subjects with anti-HBs ≥100 mIU/ml increased from 14.7% to 93.6% after the challenge dose in the DTPa-HBV-IPV/Hib group and from 19.1% to 94.4% in the DTPa-IPV/Hib+HBV group (Table 1). The post-challenge anti-HBs GMC increased by at least 187-fold in each group.

All but four initially seronegative subjects (anti-HBs <6.2 mIU/ml prior to the challenge) had an anamnestic response to vaccination (Table 1) (three in the DTPa-HBV-IPV/Hib group and one in the DTPa-IPV/Hib+HBV group). Of these subjects, two in the DTPa-HBV-IPV/Hib group and one in the DTPa-IPV/Hib+HBV group failed to develop detectable antibodies after the challenge dose. All three children with post-challenge anti-HBs <6.2 mIU/ml had post-primary anti-HBs concentrations ≥100 mIU/ml.

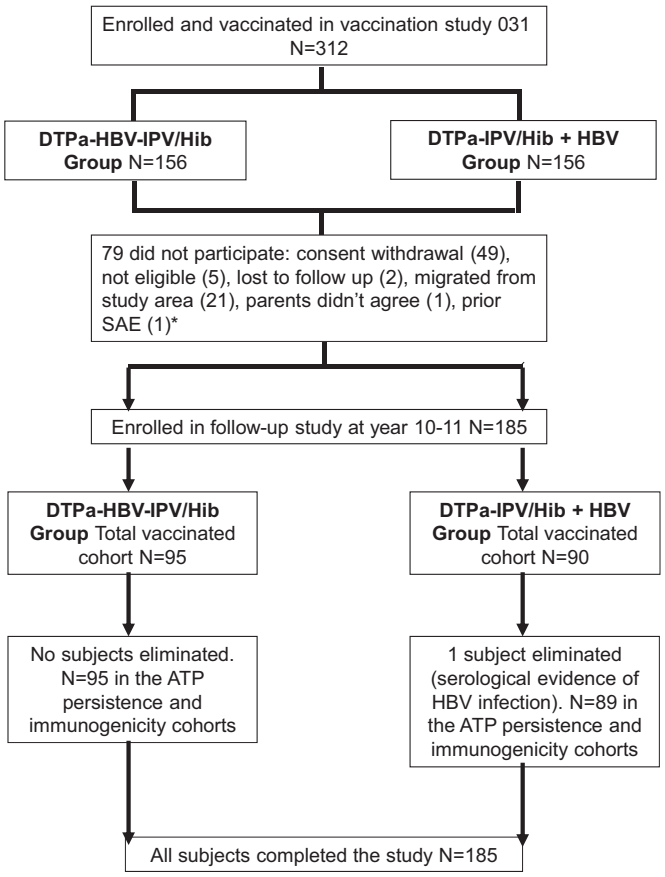


Fig. 1. Subject flow through the primary and follow-up studies.*During the primary vaccination study the subject (DTPa-HBV-IPV/Hib group) developed bilateral otitis media and developed tonic clonic spasms with epileptic crisis with onset 1 day after dose 2 and after a fall. The investigator assessed the event as unrelated to study vaccination.

All initially seropositive subjects (anti-HBs ≥6.2 mIU/ml prior to the challenge) in both groups achieved seroprotective antibody levels and all except two initially seropositive subjects in the DTPa-IPV/Hib+HBV group mounted an anamnestic response to the challenge dose (Table 1).

The kinetics of the immune response from before primary vaccination until after the challenge dose administered 10–11 years later, are given in Table 2. Children in both groups with anti-HBs concentrations <10 mIU/ml at the time of the challenge dose tended to have anti-HBs responses of a lower magnitude after each priming dose and after the booster dose than the full cohort of children who participated in the challenge study (Table 2). About 44% of the variation in response variable (post challenge dose results) is explained by the pre-challenge dose results (Fig. 2, $R^2 = 0.417$), which implies that the magnitude of the post-challenge response was not perfectly associated with the level of persisting anti-HBs antibodies prior to challenge dose.

3.2. Reactogenicity and safety

Local and general symptoms were reported by 44.2% and 31.6% of all subjects after the challenge dose. One SAE was reported (infection [unspecified] with onset 2 days after the challenge dose in a subjects in the DTPa-HBV-IPV/Hib group), but was considered by

Table 1
Antibody persistence and response to challenge (ATP cohort for immunogenicity)[*]

Pre-challenge serostatus	Timing	*N*	Post-challenge								
			≥6.2 mIU/ml		≥10 mIU/ml		≥100 mIU/ml		GMC	Anamnestic response	
			n	% (95% CI)	*n*	% (95% CI)	*n*	% (95% CI)	(95% CI)	*n*	% (95% CI)
DTPa-HBV-IPV/Hib											
<6.2 mIU/ml	Pre	42	0	0.0 (0.0; 8.4)	0	0.0 (0.0; 8.4)	0	0.0 (0.0; 8.4)	3.1 (3.1; 3.1)		–
	Post	41	39	95.1 (83.5; 99.4)	38	92.7 (80.1; 98.5)	35	85.4 (70.8;94.4)	426.5 (231.5; 785.9)	38	92.7 (80.1; 98.5)
6.2–<10 mIU/ml	Pre	7	7	100 (59.0; 100)	0	0.0 (0.0; 41.0)	0	0.0 (0.0; 41.0)	8.1 (7.1; 9.2)		–
	Post	7	7	100 (59.0; 100)	7	100 (59.0; 100)	7	100 (59.0; 100)	2775.3 (802.1; 9602.8)	7	100 (59.0; 100)
≥10 mIU/ml	Pre	46	46	100 (92.3; 100)	46	100 (92.3; 100)	14	30.4 (17.7; 45.8)	55.8 (40.1; 77.6)		–
	Post	46	46	100 (92.3; 100)	46	100 (92.3; 100)	46	100 (92.3; 100)	12180.0 (7887.9; 18807.6)	46	100 (92.3; 100)
Overall	Pre	95	53	55.8 (45.2; 66.0)	46	48.4 (38.0; 58.9)	14	14.7 (8.3; 23.5)	13.5 (9.7; 18.7)		–
	Post	94	92	97.9 (92.5; 99.7)	91	96.8 (91.0; 99.3)	88	93.6 (86.6; 97.6)	2528.6 (1572.1; 4067.1)	91	96.8 (91.0; 99.3)
DTPa-IPV/Hib + HBV											
<6.2 mIU/ml	Pre	32	0	0.0 (0.0; 10.9)	0	0.0 (0.0; 10.9)	0	0.0 (0.0; 10.9)	3.1 (3.1; 3.1)		–
	Post	32	31	96.9 (83.8; 99.9)	31	96.9 (83.8; 99.9)	27	84.4 (67.2; 94.7)	919.7 (432.3; 1956.6)	31	96.9 (83.8; 99.9)
6.2–<10 mIU/ml	Pre	5	5	100 (47.8; 100)	0	0.0 (0.0; 52.2)	0	0.0 (0.0; 52.2)	7.1 (5.5; 9.2)		–
	Post	5	5	100 (47.8; 100)	5	100 (47.8; 100)	17	100 (47.8; 100)	3660.9 (755.2; 17746.5)	5	100 (47.8; 100)
≥10 mIU/ml	Pre	52	52	100 (93.2; 100)	52	100 (93.2; 100)	17	32.7 (20.3;47.1)	56.8 (40.1; 80.5)		–
	Post	52	52	100 (93.2; 100)	52	100 (93.2; 100)	52	100 (93.2; 100)	9192.8 (5729.1; 14750.8)	50	96.2 (86.8; 99.5)
Overall	Pre	89	57	64.0 (53.2; 73.9)	52	58.4 (47.5; 68.8)	17	19.1 (11.5; 28.8)	17.8 (12.4; 25.4)		–
	Post	89	88	98.9 (93.9; 100)	88	98.9 (93.9; 100)	84	94.4 (87.4; 98.2)	3815.1 (2442.3; 5959.5)	86	96.6 (90.5; 99.3)

N = number of subjects with available CLIA results; *n* (%) = number (percentage) of subjects with concentration within the specified range. 95% CI = 95% confidence interval; GMC = geometric mean antibody concentration calculated on all subjects; Pre = pre-challenge dose; Post = 1 month post-challenge dose.

[*] Note that the ATP immunogenicity cohort was identical to the ATP persistence cohort.

Table 2
Anti-HBs antibodies from pre-primary vaccination to post-challenge in all subjects, and in subjects with anti-HBs antibodies <10 mIU/ml prior to challenge (ATP cohort for immunogenicity).

Timing	*N*	≥10 mIU/ml		≥100 mIU/ml		GMC
		n	% (95% CI)	*n*	% (95% CI)	Value (95% CI)
All subjects in the challenge phase						
DTPa-HBV-IPV/Hib						
Pre	95	9	9.5 (4.4; 17.2)	5	5.3 (1.7; 11.9)	7.1 (5.6; 9.1)
Post-dose 2	94	90	95.7 (89.5; 98.8)	84	89.4 (81.3; 94.8)	562.3 (406.1; 778.6)
Pre-dose 3	95	89	93.7 (86.8; 97.6)	67	70.5 (60.3; 79.4)	158.1 (120.0; 208.3)
Post-dose 3	95	93	97.9 (92.6; 99.7)	93	97.9 (92.6; 99.7)	3799.7 (2767.1; 5217.5)
Pre-challenge	95	46	48.4 (38.0; 58.9)	14	14.7 (8.3; 23.5)	13.5 (9.7; 18.7)
Post-challenge	95	91	96.8 (91.0; 99.3)	88	93.6 (86.6; 97.6)	2528.6 (1572.1; 4067.1)
DTPa-IPV/Hib+HBV						
Pre	89	7	7.9 (3.2; 15.5)	5	5.6 (1.8; 12.6)	6.9 (5.3; 8.9)
Post-dose 2	89	75	84.3 (75.0; 91.1)	47	52.8 (41.9; 63.5)	88.0 (60.3; 128.5)
Pre-dose 3	88	85	96.6 (90.4; 99.3)	49	55.7 (44.7; 66.3)	111.9 (84.0; 149.3)
Post-dose 3	89	89	100 (95.9; 100)	89	100 (95.9; 100)	3177.7 (2504.8; 4031.4)
Pre-challenge	89	52	58.4 (47.5; 68.8)	17	19.1 (11.5; 28.8)	17.8 (12.4; 25.4)
Post-challenge	89	88	98.9 (93.9; 100)	84	94.4 (87.4; 98.2)	3815.1 (2442.3; 5959.5)
Subjects with pre-challenge anti-HBs <10 mIU/ml						
DTPa-HBV-IPV/Hib						
Pre	49	6	12.2 (4.6; 24.8)	2	4.1 (0.5; 14.0)	7.4 (5.3; 10.3)
Post-dose 2	49	45	91.8 (80.4; 97.7)	39	79.6 (65.7; 89.8)	310.1 (185.5; 518.4)
Pre-dose 3	49	43	87.8 (75.2; 95.4)	25	51.0 (36.3; 65.6)	81.7 (54.5; 122.4)
Post-dose 3	49	47	95.9 (86.0; 99.5)	47	95.9 (86.0; 99.5)	1655.3 (1040.1; 2634.3)
Pre-challenge	49	0	0.0 (0.0; 7.3)	0	0.0 (0.0; 7.3)	3.6 (3.2; 3.9)
Post-challenge	48	45	93.8 (82.8; 98.7)	42	87.5 (74.8; 95.3)	560.5 (316.6; 992.1)
DTPa-HBV-IPV+Hib						
Pre	37	2	5.4 (0.7; 18.2)	2	5.4 (0.7; 18.2)	6.3 (4.6; 8.6)
Post-dose 2	37	28	75.7 (58.8; 88.2)	13	35.1 (20.2; 52.5)	44.6 (24.9; 79.8)
Pre-dose 3	37	34	91.9 (78.1; 98.3)	13	35.1 (20.2; 52.5)	49.9 (33.1; 75.5)
Post-dose 3	37	37	100 (90.5; 100)	37	100 (90.5; 100)	1414.4 (1020.6; 1960.1)
Pre-challenge	37	0	0.0 (0.0; 9.5)	0	0.0 (0.0; 9.5)	3.5 (3.1; 3.8)
Post-challenge	37	36	97.3 (85.8; 99.9)	32	86.5 (71.2; 95.5)	1108.5 (560.4; 2192.4)

Pre, just before the first vaccine dose at Day 0 time point; Post-dose 2, 1 month after second vaccine dose at Month 3 time point; Pre-dose 3, just before the third vaccine dose at Month 8 time point; Post-dose 3, 1 month after the third dose; Pre-challenge, Pre challenge dose time point; Post-challenge, Post challenge dose time point; GMC, geometric mean antibody concentration calculated on all subjects; *N*, number of subjects with available results; *n*/%, number/percentage of subjects with concentration within the specified range; 95% CI, 95% confidence interval.
Note: Samples from pre-primary vaccination until post -dose 3 were tested using an enzyme-linked immunosorbent assay. The pre-challenge and the post-challenge results were measured with a validated CLIA.

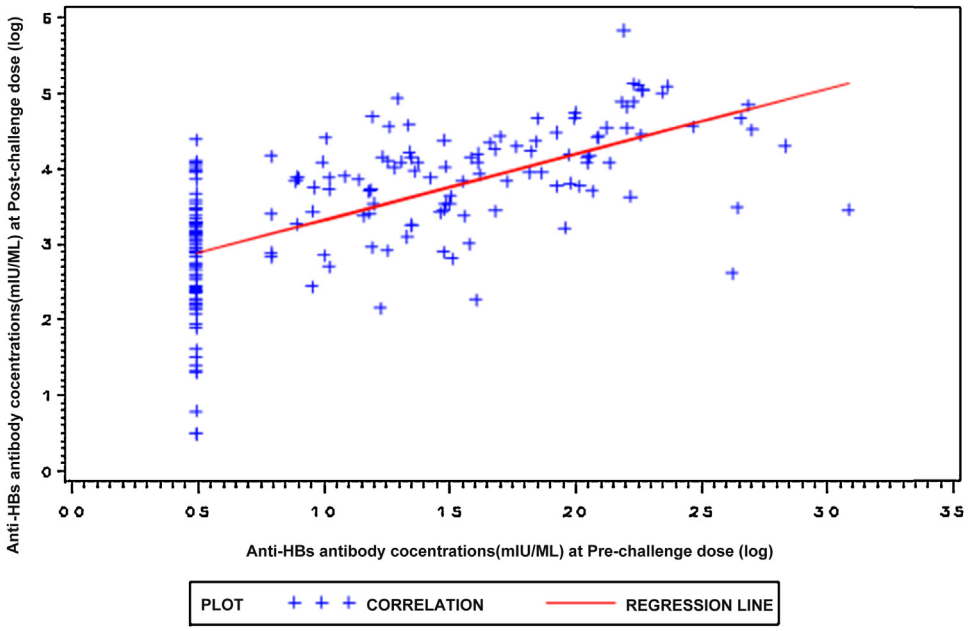


Fig. 2. Anti-HBs antibody concentrations post-challenge as a function of pre-challenge concentrations, with regression line (ATP Immunogenicity cohort). $R^2 = 0.417$. R^2 = proportion of variation in post-challenge dose (log) that is predictable from pre-challenge dose (log).

the investigator to be unrelated to vaccination. The event resolved during the study period.

4. Discussion

To our knowledge this is first study to show that combined DTPa-HBV-IPV/Hib vaccine induced sustained immune memory to HBV 10–11 years after vaccination at 3, 5 and 11–12 months of age, demonstrated by a marked increase in GMC and in the percentage of subjects who reached the 100 mIU/ml antibody threshold after the challenge HBV dose. At year 10–11, over 50% of subjects in each group still had detectable anti-HBs antibodies. Importantly, most of those that were seronegative (over 90%) demonstrated an anamnestic response, with anti-HBs antibody levels ≥10 mIU/ml after the HBV challenge dose. Overall, the data from our study suggest that protection against HBV may still be conferred through immune memory in the majority of subjects who responded to the primary vaccination, but whose anti-HBs antibody concentrations fell below 10 mIU/ml over time.

Two studies in Italy that assessed long term anti-HBs antibody persistence in children vaccinated with DTPa-HBV-IPV/Hib using the 3, 5, 11–12 months schedule showed that up to 5 years after vaccination with DTPa-HBV-IPV/Hib in infancy, 83.2% of children continued to have anti-HBs antibodies ≥10 mIU/ml [19,20]. Our study 10–11 years after vaccination showed that anti-HBs concentrations ≥10 mIU/ml were maintained in 48.4% of DTPa-HBV-IPV/Hib vaccinees and 58.4% of DTPa-IPV/Hib + HBV vaccinees, suggesting waning of antibody levels over time. In Italian subjects boosted with monovalent HBV, anamnestic responses were observed in 94.3% of children who were anti-HBs seronegative 5 years after priming [20].

The data from our persistence study of 3, 5, 11–12 month-primed individuals are in line with other published reports following vaccination of infants with 3 priming HBV vaccine doses and no booster, or 3 priming doses with second year of life booster, including studies of antibody persistence after primary vaccination with *Infanrix hexa*™ [13,15,16,21–23]. Subjects in these studies showed anamnestic responses to challenge up to 20 years after vaccination, despite the absence of circulating antibody. The results of the present study indicate that infant vaccination using the 3, 5, 11–12 month schedule appears to be similarly effective in conferring lasting immunity against HBV compared to 3-dose priming schedules. The data indicate that current recommendations of WHO and the EU Consensus Group on Hepatitis B Immunity are also applicable to infants vaccinated at 3, 5, 11–12 months of age [17,18].

We observed no differences between HBV vaccine administered separately with DTPa-IPV/Hib or combined as DTPa-HBV-IPV/Hib in anti-HBs persistence or in the response to the challenge dose, suggesting that long term protection against HBV afforded by the combined DTPa-HBV-IPV/Hib vaccine is the similar to that provided by monovalent *Engerix*™ B. The monovalent vaccine has been shown to induce long term protection and immune memory up to 20 years post-vaccination [23].

We observed that the magnitude of the response to challenge was only moderately linked to the magnitude of the pre-challenge anti-HBs antibody concentration. However, there was a trend for lower post-primary and post-challenge anti-HBs concentrations in subjects with pre-challenge anti-HBs concentrations <10 mIU/ml. The small number of subjects in this study does not allow statistical comparisons or conclusions to be drawn, although other studies have noted correlations between post-primary, pre-challenge and post-challenge anti-HBs levels [13,15,24,25]. There is growing evidence that immune memory to HBV is primarily driven by persisting memory cells rather than by persisting circulating antibody [16,26]. Thus, the clinical relevance of potential correlations between post-primary, pre-challenge and post-challenge anti-HBs levels is uncertain.

The incidence of local and general symptoms after the challenge dose was similar regardless of the primary vaccination received, and in line with the known safety profile of *Engerix*™ B.

DTPa-HBV-IPV/Hib immunizes against 6 common diseases of childhood in a single injection. This study builds on previous studies showing that immune memory to HBV induced after primary vaccination with DTPa-HBV-IPV/Hib is as good as that induced by HBV vaccine alone. Furthermore, administration of DTPa-HBV-IPV/Hib or HBV at 3, 5 and 11–12 months of age resulted in strong anamnestic responses to HBV 10–11 years after vaccination in the majority of children.

Contributors

M.A. was the principal investigator and together with K.H. and M.K. contributed to the conception, design, analysis and interpretation of the study. P.D.C. undertook the statistical analysis. All authors participated in the development of this manuscript.

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Conflicts of interest

P.C., K.H. and M.K. are employees of GlaxoSmithKline Biologicals SA and K.H. and M.K. have stock options from the sponsoring company. M.A. has declared no conflict of interest.

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